

Project – Electric Vehicle Population Data Analysis

Introduction:

As the automotive industry rapidly transitions towards sustainability, understanding the adoption patterns and infrastructure landscape of electric vehicles (EVs) is critical. The Electric Vehicle Analysis Dashboard provides an in-depth overview of EV distribution across U.S. states, model years, makes, and types (BEV & PHEV). This interactive dashboard empowers analysts, policymakers, and automakers to identify trends in electric mobility, understand manufacturer performance, and evaluate vehicle eligibility for clean fuel programs like CAFV (Clean Alternative Fuel Vehicle). This dashboard combines KPIs, geographic heatmaps, and trend charts to reveal how electric vehicle technology has evolved over time and where it is most prevalent.

Steps:

1. Load Data:

Loaded data by creating connection with 'Electric_Vehicle_Population_Data.csv'

2. Sheet 1: Total Vehicles

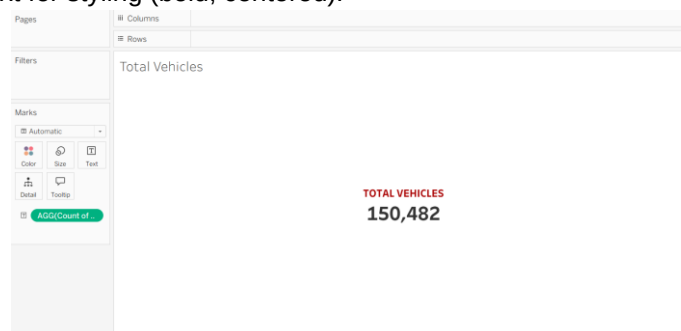
Objective: Displays a KPI card with the calculated total vehicle count.

2.1. Created a calculated field 'Count of Vehicles':



2.2. Dragged Count of Vehicles to Text → AGG(Count of Vehicles)

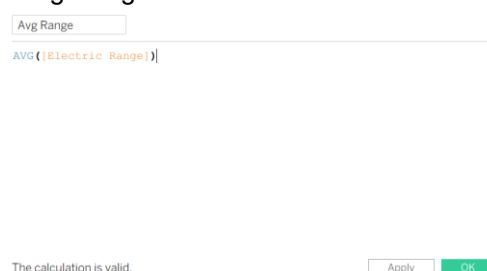
2.3. Formatted the text for styling (bold, centered).



3. Sheet 2: Average Electric Range

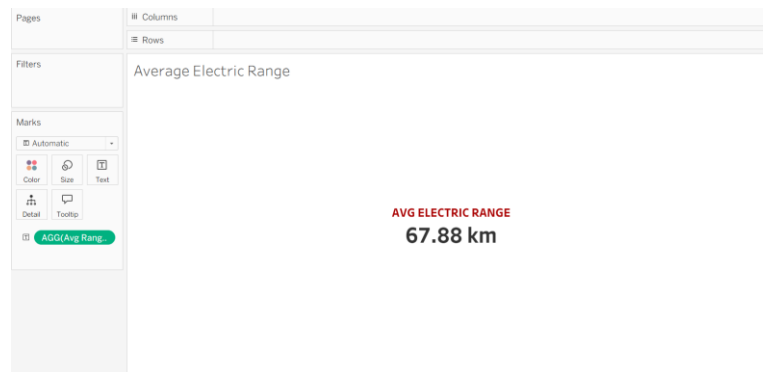
Objective: Display the average electric range (in kilometers) across all EVs.

3.1. Created a calculated field 'Avg Range':



3.2. Dragged 'Avg Range' to Title → AGG('Avg Range')

3.3. Formatted the text for styling (bold, centered)



4. Sheet 3: BEV Vehicles

Objective: Highlight how many vehicles are Battery Electric Vehicles and what percentage they constitute.

4.1. Created a calculated field 'BEV Vehicles':

```
BEV Vehicles

Count (
  IF [Electric Vehicle Type]='Battery Electric Vehicle (BEV) '
  THEN [Vin (1-10)]
  END
)
```

The calculation is valid.

Apply OK

4.2. Created another calculated field '% of BEV':

```
% of BEV

[BEV Vehicles]/[Count of Vehicles]
```

The calculation is valid.

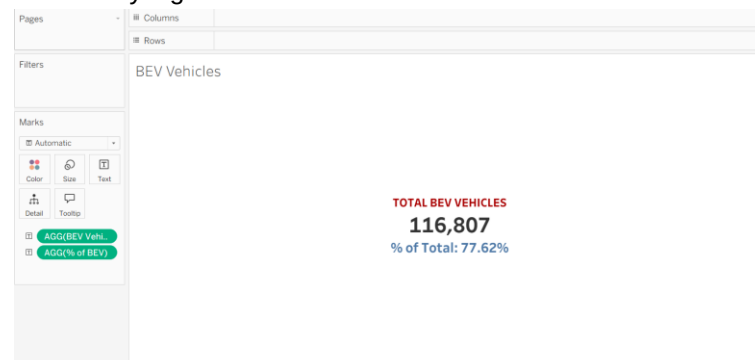
1 Dependency

Apply OK

4.3. Dragged 'BEV Vehicles' to Title → AGG(BEV Vehicles)

4.4. Dragged '% of BEV' to Title → AGG(% of BEV) → formatted as percentage

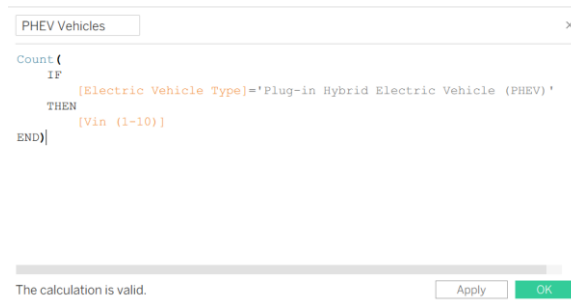
4.5. Formatted the text for styling



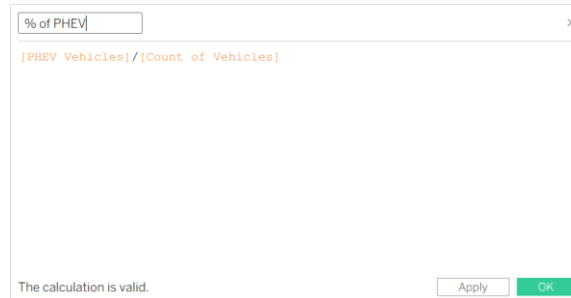
5. Sheet 4: PHEV Vehicles

Objective: Show how many Plug-in Hybrid Electric Vehicles are present and their percentage.

5.1. Created a calculated field 'PHEV Vehicles':



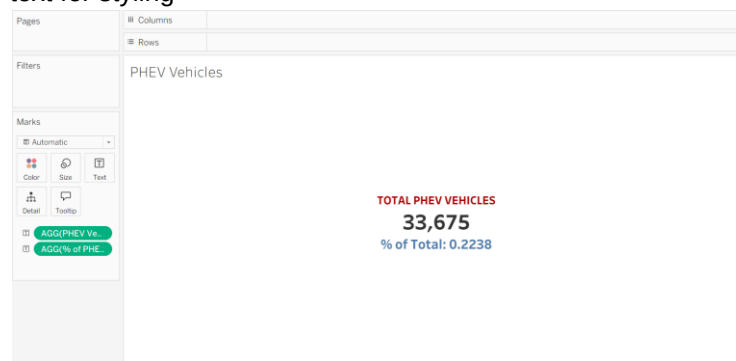
5.2. Created another calculated field ' % of PHEV':



5.3. Dragged 'PHEV Vehicles' to Title → AGG(PHEV Vehicles)

5.4. Dragged '% of PHEV' to Title → AGG(% of PHEV) → formatted as percentage

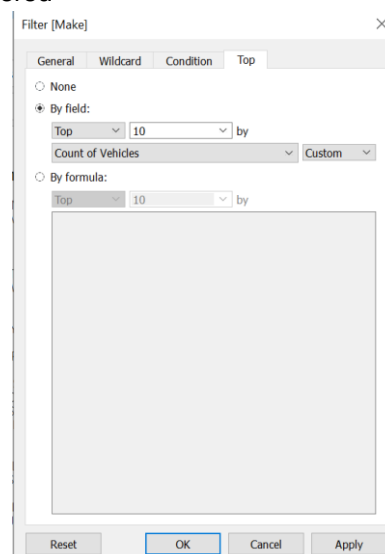
5.5. Formatted the text for styling



6. Sheet 5: Top 10 # of Vehicles by Make

Objective: Identify the top 10 manufacturers by EV count

6.1. Used 'Make' on Rows and filtered



6.2. 'Count of Vehicles' on Columns → AGG(Count of Vehicles)

6.3. Horizontal bar chart, sorted descending.

6.4. Added 'Model Year' in Filters:

Filter [Model Year]

Range of values | **At least** | At most | Special

At least

2,011 | 2,024

1997 | 2024

Show: Only Relevant Values | Include Null Values

Reset | OK | Cancel | Apply

6.5. Create a parameter 'Top N':

Create Parameter

Name
Top N

Properties

Data type: Integer | Display format: 1

Current value: 1 | Value when workbook opens: Current value

Allowable values
☐ All ☐ List ☒ Range

Range of values

☒ Minimum: 1 ☒ Maximum: 50 ☐ Step size: 1

☒ Fixed ☐ When workbook opens

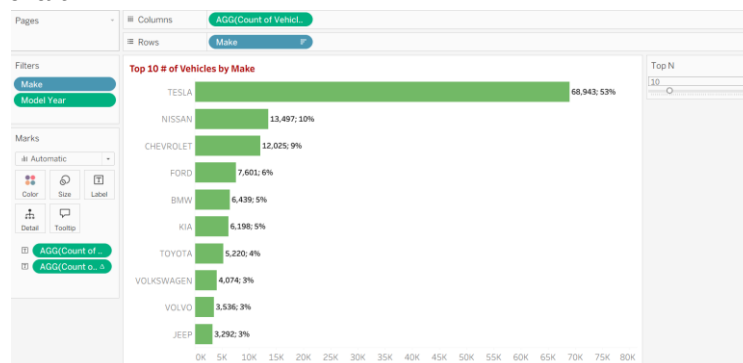
Add values from

Cancel | OK

6.6. Set current value of 'Top N' parameter to 10 → Show Parameter

6.7. Edited Title → Top <Parameters.Top N> # of Vehicles by Make

6.8. Formatted the chart



7. Sheet 6: # of Vehicles by State

Objective: Display the geographical distribution of EVs across U.S. states.

7.1. Dragged 'Latitude' to Rows

7.2. Dragged 'Longitude' to Columns

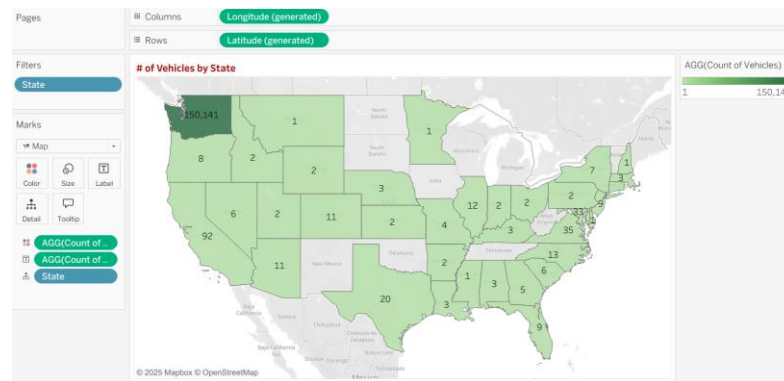
7.3. Used 'State' as Details on map.

7.4. Dragged 'Count of Vehicles' in Label and Color

7.5. Set Marks as Map

7.6. Excluded data of outside USA

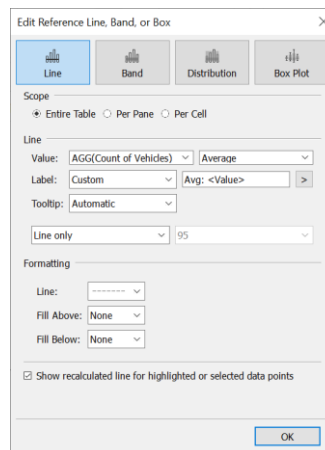
7.7. Filtered State to remove Null values



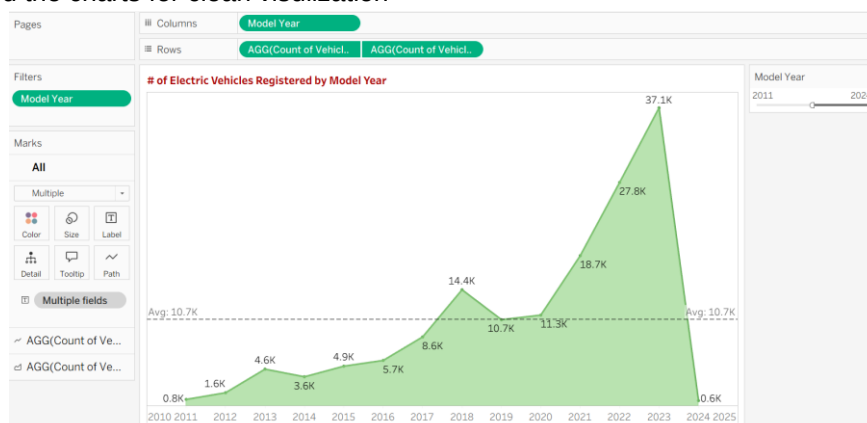
8. Sheet 7: # of Vehicles Registered by Model Year

Objective: Show the trend of EV adoption over time.

- 8.1. Dragged 'Model Year' to Columns and 'Count of Vehicles' to Rows.
- 8.2. Duplicated AGG(Count of Vehicles) → Dual Axis
- 8.3. Set one Marks as Line and another as Area.
- 8.4. Synchronized the axes, and formatted for clarity
- 8.5. Dragged 'Count of Vehicles' in the Color
- 8.6. From 'Analytics', dragged 'Average Line' and edited:



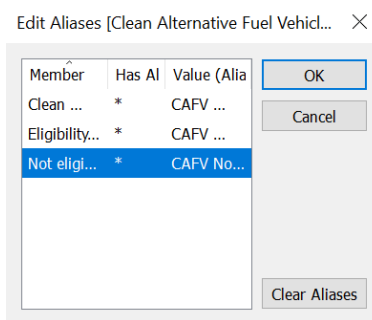
- 8.7. Dragged 'Model Year' to Filters for setting starting year as 2011
- 8.8. Formatted the charts for clean visualization



9. Sheet 8: Vehicles Eligibility for CAFV

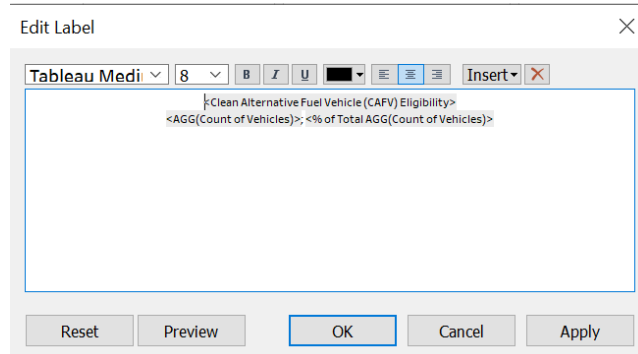
Objective: Visualize the proportion of EVs eligible for clean fuel programs

- 9.1. Double clicked in Rows → AVG(0)
- 9.2. Marks → Pie
- 9.3. Edited Aliases of 'Clean Alternative Fuel Vehicle (CAFV) Eligibility'



9.4. Dragged 'Clean Alternative Fuel Vehicle (CAFV) Eligibility' to Color and Label

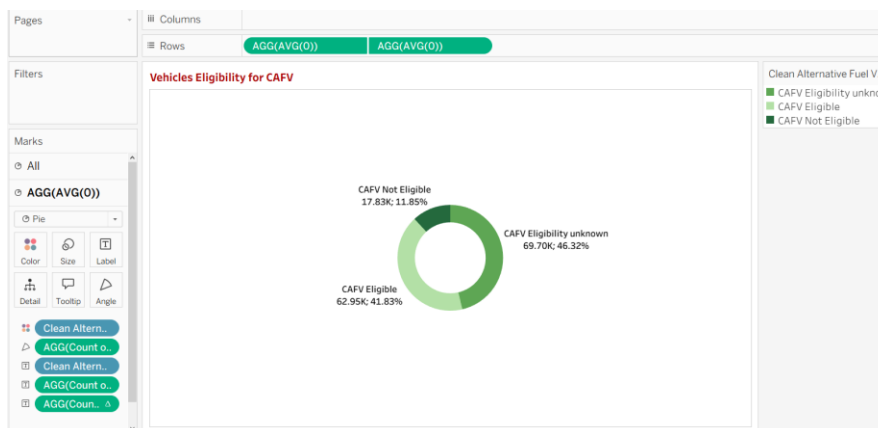
9.5. Dragged 'Count of Vehicles' to Angle and Label → Percent of Total. Dragged 'Count of Vehicles' to Label again. Formatted the label



9.6. Ctrl + dragged AGG(AVG(0)) in Rows → Dual Axis

9.7. Removed all values from 2nd Pie, set Color as white, updated Size

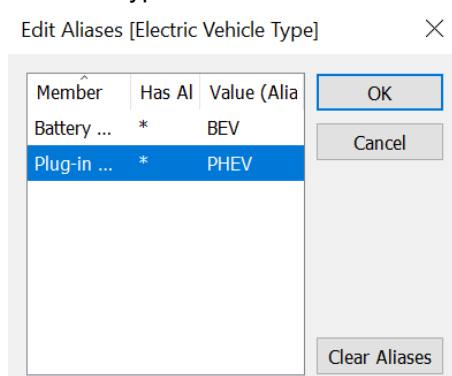
9.8. Formatted for better visualization.



10. Sheet 9: # of Vehicles by Model

Objective: Show top models by count, along with their type (BEV/PHEV) and make.

10.1. Edited Aliases of 'Electric Vehicle Type'

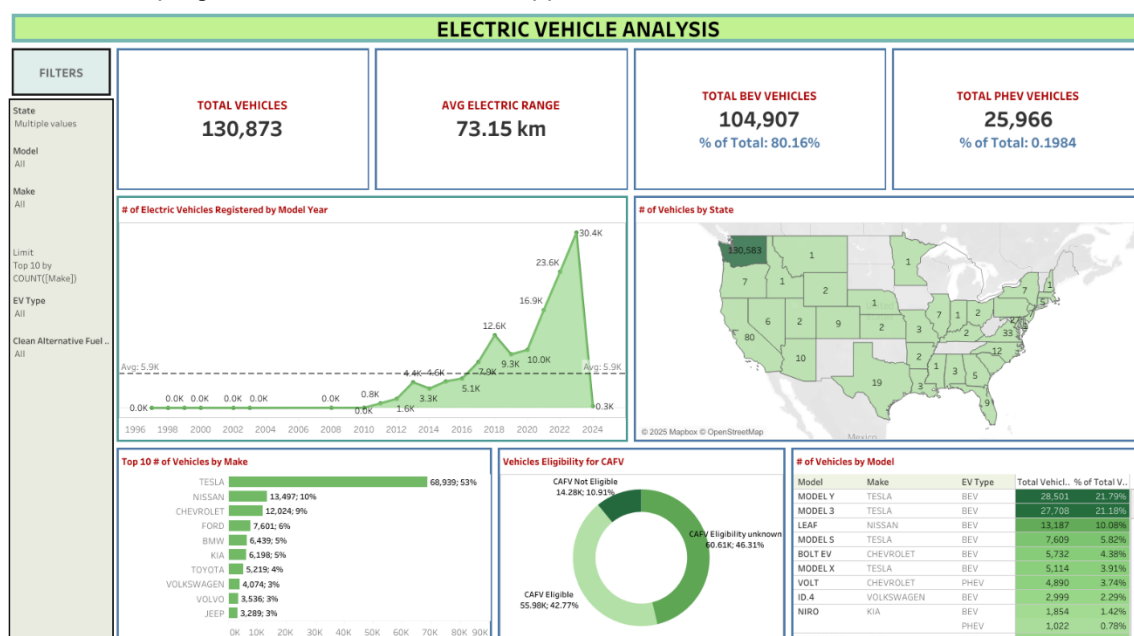


- 10.2. Added Make, Model, EV Type to Rows
- 10.3. Added 'Count of Vehicles' in Color
- 10.4. Set Marks as Square
- 10.5. Added Measure Values 'Count of Vehicles' → Percent of Total
- 10.6. Added Measure Values 'Count of Vehicles'
- 10.7. Sorted and Formatted the Color
- 10.8. Added required filters applied to all using the same data source.

Model	Make	EV Type	Total Vehicles	% of Total Vehicles
MODEL Y	TESLA	BEV	28,501	21.79%
MODEL 3	TESLA	BEV	27,708	21.18%
LEAF	NISSAN	BEV	13,187	10.08%
MODEL S	TESLA	BEV	7,609	5.82%
BOLT EV	CHEVROLET	BEV	5,732	4.38%
MODEL X	TESLA	BEV	5,114	3.91%
VOLT	CHEVROLET	PHEV	4,890	3.74%
ID.4	VOLKSWAGEN	BEV	2,999	2.29%
NIRO	KIA	BEV	1,854	1.42%
WRANGLER	JEEP	PHEV	1,022	0.78%
MUSTANG	FORD	BEV	2,626	2.01%
PRIUS PR	TOYOTA	PHEV	2,527	1.93%
XS	BMW	PHEV	2,068	1.58%
I3	BMW	BEV	592	0.45%
FUSION	FORD	PHEV	1,325	1.01%
EV6	KIA	BEV	1,829	1.40%
CAMAX	FORD	PHEV	1,644	1.26%
RAV4 PRL	TOYOTA	PHEV	1,546	1.18%
XC90	VOLVO	PHEV	1,525	1.17%
BOLT EUV	CHEVROLET	BEV	1,233	0.94%
E-GOLF	VOLKSWAGEN	BEV	1,153	0.88%
			1,075	0.82%

11. Dashboard

- 11.1. Set customize size 1600X900
- 11.2. Add horizontal and vertical containers for layout.
- 11.3. Place KPIs as floating elements with appropriate titles and formatting.
- 11.4. Insert maps, and other charts into fixed dashboard positions.
- 11.5. Added Filters in the Filter Panel
- 11.6. Apply final formatting such as removing gridlines, adding borders, and color coding KPIs and progress bars for better visual appeal.



Key Findings:

1. **Total Vehicles:**
The dataset includes **130,873** electric vehicles, with a majority being BEVs.
2. **Vehicle Type:**
 - 80.16% are **Battery Electric Vehicles (BEV)**.
 - 19.84% are **Plug-in Hybrid Electric Vehicles (PHEV)**.
3. **Top Manufacturers:**
 - TESLA** dominates with **68,939** vehicles (53%).
 - NISSAN and CHEVROLET follow with 13,497 and 12,024 respectively.

4. **Top Models:**
 - **Model Y** and **Model 3** from **TESLA** are the most popular.
 - **LEAF** (Nissan) and **Bolt EV** (Chevrolet) are notable BEV contenders.
5. **CAFV Eligibility:**
 - **46.31%** of vehicles have **unknown eligibility**.
 - Only **42.77%** are marked as **CAFV Eligible**.
 - **10.91%** are **not eligible**, suggesting data gaps or policy exclusions.
6. **Adoption Over Time:**
 - EV registrations grew significantly **post-2015**, peaking in **2023 with 30.4K registrations**.
 - A minor drop is noted in 2024, possibly due to incomplete data or market saturation.
7. **State-Level Distribution:**
 - **Washington State (130,583)** leads by a wide margin.
 - Other states show minimal adoption, highlighting geographic disparity in EV presence.
8. **Average Range:**
 - The **average electric range is 73.15 km**, suggesting the dataset may include older models or low-range PHEVs.

Conclusion:

The **Electric Vehicle Analysis Dashboard** presents a comprehensive overview of the electric vehicle landscape in the U.S., offering valuable insights into adoption trends, market dominance, and clean energy compliance. The overwhelming dominance of BEVs and the significant lead of Tesla indicate market consolidation around a few key players.

This dashboard can aid **transport authorities**, **auto manufacturers**, and **clean energy advocates** in making **data-backed decisions**, developing **targeted policies**, and planning **infrastructure rollouts** to support a more equitable and widespread EV future.